Review 05 — Pulsed × annular (new science)

`mcdo_dev` Phase 5 (`output/05_pulsed_pupils/`, `docs/theory/05_pulsed_pupils.md`)

2026-06-13

The progress report's "immediate next step"

Eq.-10 spectral sum (full spectrum, Phase 1) over annular fields (Babinet, Phase 3). $X_{NA} = 0.8$, λ_c=750 nm, linear-x y-cut, N_freq=201.

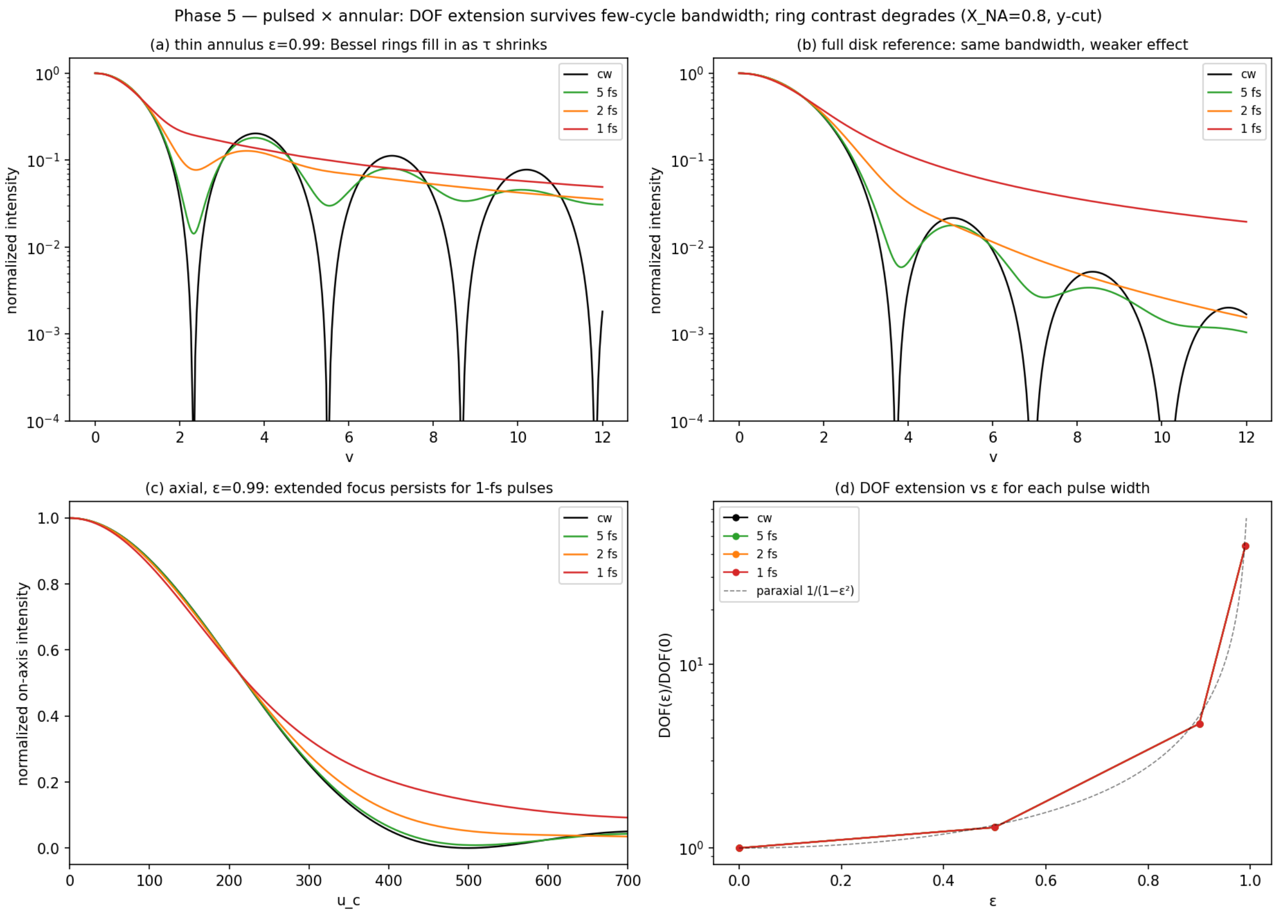
(i) Annular DOF extension is immune to few-cycle bandwidth.

DOF(ε)/DOF(0)	cw	5 fs	2 fs	1 fs
$\varepsilon = 0.50$	1.30	1.30	1.30	1.30
$\varepsilon = 0.90$	4.76	4.76	4.76	4.76
$\varepsilon = 0.99$	44.35	44.35	44.34	44.33

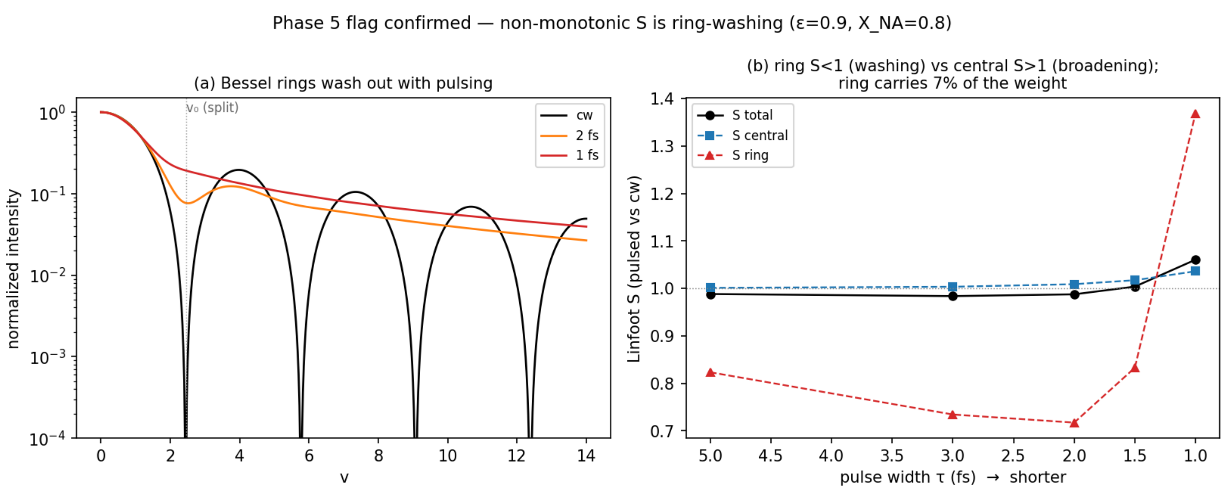
Invariant to 3 digits down to 1 fs (geometric $1/(1-\varepsilon^2)$, NA fixed across frequencies). Each configuration's own DOF lengthens only ∼1.0% cw→1 fs.

(ii) Transverse ring contrast erodes with pulsing, as the Romallosa zeros did: Linfoot $S(1\,\text{fs}) \to 1.058$ across ε. Cross-check: $\varepsilon = 0$ gives $S(1\,\text{fs}) = 1.056$, matching Phase-1 single-photon (1.053–1.062) — pipeline confirmed.

Flag: $S(\tau)$ is *non-monotonic* at high ε ($0.988 \to 0.987 \to 1.058$ for 5/2/1 fs) — moderate pulses wash out the Bessel rings ($S < 1$) before 1-fs central broadening dominates ($S > 1$). Plausible ring-physics; S(cw,cw)=1 and the ε=0 cross-check argue against a bug; worth a dedicated ring-energy-vs-τ look if load-bearing.



(a) ε=0.99 rings fill as $\tau\downarrow$; **(b)** full disk reference; **(c)** axial extended focus persists at 1 fs; **(d)** DOF(ε)/DOF(0) curves for all τ overlie the paraxial law.

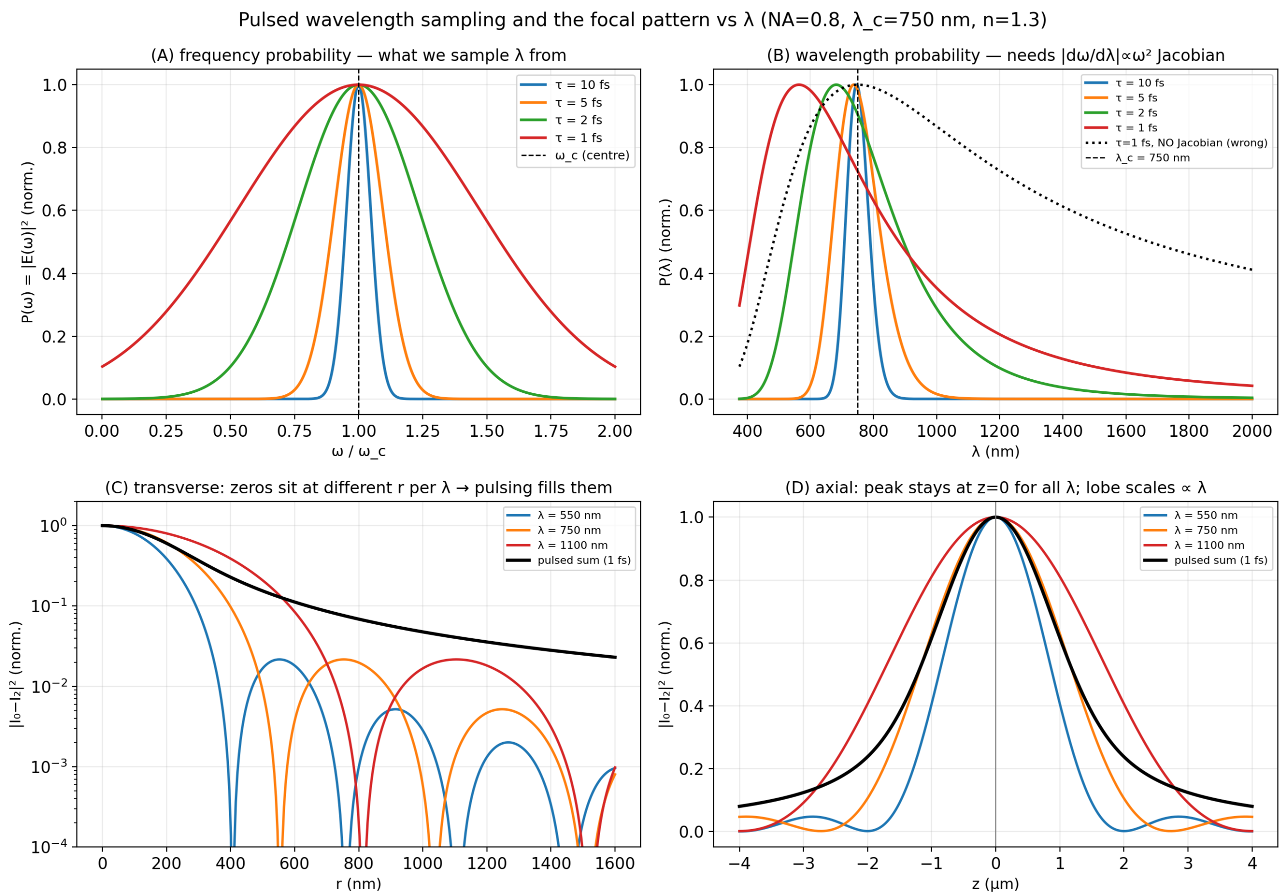


Flag confirmed. Central/ring decomposition at ε=0.9: S_central rises slowly (1.00→1.04, robust Bessel core); S_ring dips to 0.72 (rings wash out) then jumps to 1.37 at 1 fs (central skirt spills into the ring region). The rings carry only 7% of the weight, but at moderate τ S_central≈1 so the small ring deficit drags the total below 1 — hence the non-monotonicity. Not a bug (S(cw,cw)=1; ε=0 matches Phase 1).

Significance: pulsing keeps the axial DOF benefit of structured pupils, costs transverse contrast — the Romallosa trade, now for annular/Bessel pupils. This pulsed structured-pupil field is the Phase-6 (Monte Carlo) source.

Spectral sampling, and the $S(\tau)$ flag — verified

Wavelength sampling (`spectral_sampling.png`): a pulse draws λ from $|E(\omega)|^2$; the λ-probability needs the $|d\omega/d\lambda| \propto \omega^2$ Jacobian (without it the peak is mis-placed, panel B). The geometric focus stays at $(r{=}0, z{=}0)$ for *every* λ; only the pattern *scale* $\propto \lambda$ changes (zeros move), so summing over the spectrum fills the zeros — the Romallosa effect (panels C, D).



The $S(\tau)$ non-monotonicity is real (`pulsed_S_diagnostic.png`), not numerics or a single-cut artifact: (A) the dip-then-rise is *smooth* on a dense τ grid; (B) S is *flat vs* N_{freq} (51–801) — not spectral undersampling; (C) it appears in the y-cut $|I_0 - I_2|^2$, the x-cut $|I_0 + I_2|^2 + 4|I_1|^2$, *and* the full azimuth-averaged $|I_0|^2 + |I_2|^2 + 2|I_1|^2$ (all of E_x, E_y, E_z). Cross-check S(5/2/1 fs)=0.988/0.987/1.057 reproduces the table above.